

Bear with me here. Imagine that markets then continue to decline for another 19 months and your portfolio's investment return hits -10% annualized (which is -4 basis points per day). See (the dark solid line in) Figure 6.2 for an illustration of the evolution of this entire cycle from start to finish over 2π (which equals 6.28) years. This sine wave I have constructed exhibits the amplitude (volatility) of plus or minus 15% , and ranges in value from -10% to $+20\%$.

As a perfectly symmetric alternative, consider the scenario, represented by the dashed line, in which you retire and start to withdraw $\$7$ annually, while the market is earning the same 5% per annum, but it immediately moves into a bear-market cycle so that 19 months into retirement you are earning -10% (that is, losing money) per annum, and 19 months later you are back to 5% per annum and 19 months after that you are earning 20% , and so on.

Once again, the two diametrically opposed paths are illustrated in Figure 6.2. All paths earn an average APR of 5% over the long run.

Again, in the absence of any withdrawals, if you only invested $\$1$ at the start of a cycle (up or down, bullish or bearish) then at the end of a market cycle of 2π (which equals 6.28) years, you would have a compound annual return of 5% and the same amount of money: $\$1.37$ in all cases. When you are buying and holding it doesn't matter what path the market takes as long as you get the 5% annualized return! Whether you move up first and then down, or down first and then up, if your compound return is 5% , you will know exactly where you end up. But the situation is very different when you withdraw money.

I trust you would agree that on any given day, week, or month, you don't really know in which direction the market will move over the next few years. It might go north or it might go south. The relevant question is how long your portfolio will last based on the possible starting market cycles. It should be intuitive that your retirement will be worse off if markets go south versus north. But how bad will it be? That's my second point.

Table 6.1 summarizes the main results for cases in which you start retirement with $\$100$ and withdraw $\$6$, $\$7$, or $\$8$ per year on a continuous basis. For example, as you saw in Figure 6.2, if you are withdrawing $\$7$ per year and you are invested in a relatively volatile asset class that fluctuates between $+20\%$ and -10% , the portfolio can last